Chapter-5

Linear System

$$5x + 3y - 13z = 12$$
$$2x - 3y + 12z = 20$$
$$x - 2y - 17z = 21$$

Num of equation = num of unknowns.

Power of each unknown = 1

এদের corresponding matrix ও shift করে পাই।

$$\underbrace{\begin{bmatrix} 5 & 3 & -13 \\ 2 & -3 & 12 \\ 1 & -2 & -17 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 12 \\ 20 \\ 21 \end{bmatrix}}_{b}$$

So, $Ax = b$

$\therefore x = A^{-1}b = \frac{1}{\det(A)} A^{T} \cdot b$

if $\det(A) \neq 0$; then Solution Possible.

Upper Triangular Matrix

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ 0 & b_2 & b_3 \\ 0 & 0 & c_3 \end{bmatrix}$$

Lower Triangular Matrix

$$\begin{bmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & 0 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

If,

(i) Num. of eqn < Num. of unknown; Solution Not Possible

(ii) Num of eqn > Num. of unknown; Multiple Type solution

(iii) Num of eqn = Num. of unknown; Sigle solution.

Gaussian Elimination

$$\# \left[\begin{array}{cccc|c} a_{11} & 0 & 0 & 0 & b_1 \\ a_{21} & a_{22} & 0 & 0 & b_2 \\ a_{31} & a_{32} & a_{33} & 0 & b_3 \\ a_{41} & a_{42} & a_{43} & a_{44} & b_4 \end{array}\right]$$

Now,

$a_{11} x_1 = b_1$

So, $x_1 = \frac{b_1}{a_{11}}$; div = 1 ; mul = 0 ; sub = 0

$a_{21} x_1 + a_{22} x_2 = b_2$

So, $x_2 = \frac{b_2 - a_{21} x_1}{a_{22}}$; div = 1 ; mul = 1 , sub = 1

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

So, $x_3 = \dfrac{b_3 - a_{31}x_1 - a_{32}x_2}{a_{33}}$; div=1, mul=2, sub=2

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 = b_4$$

So, $x_4 = \dfrac{b_4 - a_{41}x_1 - a_{42}x_2 - a_{43}x_3}{a_{44}}$; div=1, mul=3, sub=3

So, we can say,

$x_j \Rightarrow$ 1 div, $j-1$, mul, $j-1$ sub

So, x_j requires ; $1+2(j-1)$ operation

$= 1+2j-2$ operation

$= (2j-1)$ operation.

So, total operation requires,

$$\sum_{j=1}^{n} (2j-1)$$

$$= 2\sum_{j=1}^{n} j - \sum_{j=1}^{n} 1$$

$$= 2\sum_{j=1}^{n} - n$$

$$= 2 \times \frac{n^2+n}{2} - n$$

$$= n^2 .$$

$2x_1 + 5x_2 + 9x_3 = 62$

$5x_1 - 3x_2 + 12x_3 = 56$

$2x_1 + 3x_2 - 2x_3 = 1$

$[x_1 = 1; x_2 = 3; x_3 = 5]$

Now,

$$\left[\begin{array}{ccc|c} 2 & 5 & 9 & 62 \\ 5 & -3 & 12 & 56 \\ 2 & 3 & -2 & 1 \end{array}\right]$$

First Row operation -

$R_2' = R_2 - \left(\frac{5}{2}\right) R_1$ $\quad m_{21} = 5/2$

$R_3' = R_3 - \left(\frac{3}{2}\right) R_1$ $\quad m_{31} = 3/2$

$$= \left[\begin{array}{ccc|c} 2 & 5 & 9 & 62 \\ 0 & -31/2 & -21/2 & -93 \\ 0 & -2 & -11 & -61 \end{array}\right]$$

Second Row Operation

$$R_3' = R_3 - \left(\frac{-2}{-31/2}\right) R_2$$

$$= \left[\begin{array}{ccc|c} 2 & 5 & 9 & 62 \\ 0 & -31/2 & -21/2 & -93 \\ 0 & 0 & 277/31 & -\frac{1495}{31} \end{array}\right]$$

Now,

$$\frac{-200}{31} x_3 = -\frac{1495}{31}$$

So, $x_3 = \frac{1495}{299} = 5$

$$\frac{-31}{2} x_2 - \frac{21}{2} 5 = -55$$

So, $x_2 = 3$

$$2x_1 + 5x_2 + 9x_3 = 63$$

So, $x_1 = 1$

Num of eqn → Lower / Upper → $(n-1)$ → n^{th} step needed

Row operation needed

↓

Triangular + 3 eqn

$R_2 - R_1$

$R_2 - (\frac{1}{2})R_1$

$x_1 + 2x_2 + x_3 = 0$

$x_1 - 2x + 2x_3 = 4$

$2x_1 + 12x_2 - 2x_3 = 4$

$-4R_3 - 8R_2$

$R_3 - (-\frac{8}{4})R_2$

Now,

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & -2 & 2 & 4 \\ 2 & 12 & -2 & 04 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 8 & -4 & 4 \end{array}\right] \quad [R_2' = R_2 - (\tfrac{1}{1})R_1] \quad [R_3' = R_3 - (\tfrac{2}{1})R_1]$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 0 & -2 & 12 \end{array}\right] \quad [R_3' = R_3 - (-\tfrac{8}{4})R_2]$$

Now, we can say

$x_1 + 2x_2 + x_3 = 0 \ldots\ldots (i)$

$-4x_2 + x_3 = 4 \ldots\ldots (ii)$

$-2x_3 = 12$

So, $x_3 = -6$

$$x_2 = \frac{4 - x_3}{-4} = \frac{4 - (-6)}{-4}$$

$$= -2.5$$

$x_1 = -x_3 - 2x_2 =$ 11

So, $x_1 = 11$; $x_2 = -2.5$; $x_3 = -6$ (Ans.)

The upward velocity of a rocket is given at three different times in table-1.

Time (s)	Velocity (m/s)
5	106.8
8	177.2
12	279.2

The velocity data is approximated by a polynomial as,

$$v(t) = b_1 t^2 + b_2 t + b_3 \; ; \quad 5 \le t \le 12$$

Find the value of b_1, b_2 and b_3 using the Gaussian Elimination method, Find the velocity at $t = 6, 7, 5, 9, 4$ seconds.

Sol^n

Here, $v(t) = b_1 t^2 + b_2 t + b_3$

So,

$$v(5) \Rightarrow 25b_1 + 5b_2 + b_3 = 106.8$$

$$v(8) \Rightarrow 64b_1 + 8b_2 + b_3 = 177.2$$

$$v(12) \Rightarrow 144b_1 + 12b_2 + b_3 = 279.2$$

$25R_3 - 144R_1$; $R_3 \leftarrow \frac{144}{25} R_1$ | $25R_2 - 64R_1$

$-4.8R_3 - -16.8R_2$; $R_3 - \frac{-16.8}{-4.8} R_2$ | $R_2 - \frac{64}{25} R_1$

So, Now,

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 64 & 8 & 1 & 177.2 \\ 144 & 12 & 1 & 279.2 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & -16.8 & -4.76 & -335.968 \end{array}\right] \begin{array}{l} \left[R_2' = R_2 - \left(\frac{64}{25}\right) R_1\right] \\ \left[R_3' = R_3 - \left(\frac{144}{25}\right) R_1\right] \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & 0 & 0.7 & 0.76 \end{array}\right] \quad \left[R_3' = R_3 - \left(-\frac{16.8}{-4.8}\right) R_2\right]$$

So,

$0.7 b_3 = 0.76$

$\therefore b_3 = \frac{0.76}{0.7} = \frac{38}{35}$

$-4.8 b_2 - 1.56 b_3 = -96.208$

So, $b_2 = -\frac{96.208 + \left(1.56 \times \frac{38}{35}\right)}{-4.8}$

$= \frac{827}{42}$

$25b_1 + 5b_2 + b_3 = 106.8$

$\therefore b_1 = \frac{106.8 - \frac{38}{35} - 5\left(\frac{827}{42}\right)}{25}$

$= \frac{61}{210}$

LU Decomposition

We know, $Ax = b$

$x = A^{-1}b$

$\downarrow$

$O(n^3)$

n number of eqn.

Any Matrix → Upper/Lower Triangular Matrix

now Operation $Op(n-1)$

$\downarrow O(n^2)$

Solution

Gaussian

$$\begin{bmatrix} 5 & 3 & 2 \\ 3 & 2 & 9 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} 27 \\ 7 \\ 2 \end{bmatrix}$$

→ এক্ষেত্রে ডান right hand side may change, in this case we need to redo the calculation in Gaussian

$\downarrow$

$$\begin{bmatrix} 5 & 3 & 2 \\ 0 & 1/5 & 29/5 \\ 0 & 0 & -85 \end{bmatrix} \begin{bmatrix} 27 \\ -46/5 \\ 107 \end{bmatrix}$$

যাতে আমাদের calculation বারবার করে বোঝার না করার জন্যে তাই আমরা procedure use করি এটাই LU Decomposition

$Ax = b$ 로 표현

L U ⟶ Upper Triangular Matrix

↙

Lower Triangular Matrix

Decomposition → Breakdown

$Ax = b$; $A = LU$

So, $LUx = b$ ①

$Ly = b$ ②

$Ux = y$ ③

$x + 2y + z = 0$

$x - 2y + 2z = 4$

$2x + 12y - 2z = 4$

Soln $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix}$ $b = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$

A Decomposition $A = LU$

$A = \begin{bmatrix} 1 & 2 & 1 \\ ① & -2 & 2 \\ ② & 12 & -2 \end{bmatrix}$

Part C

Frobenious Matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix}$$

$$F^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$



Now, $A^{(2)} = F^{(1)} \times A^{(1)}$

Upper Triangular Matrix

1st কৃত্রিম Create করতে

$$\begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix}$$

m এর value দিয়ে populate

যেখানে নিচেরটা 0 হবে, sign (-) হবে,

$R_2' = R_2 - \left(\frac{1}{1}\right) R_1$

$R_3' = R_3 - \left(\frac{2}{1}\right) R_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ -2 & 1 & 1 \end{bmatrix}$$

যেটা 0 হবে না, as it is নিচের 0 থাকবে

→ Multiplier এর Negative value থাকবে.

Here,

$$A^{(1)} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix}$$

$R_2' = R_2 - \left(\frac{1}{1}\right) R_1$

$R_3' = R_3 - \left(\frac{2}{1}\right) R_1$

Now, $F^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

Now,

$A^{(2)} = F^{(1)} \times A^{(1)}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{bmatrix}$$

$$A^{(2)} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 8 & -4 \end{bmatrix}$$

$R_3' = R_3 - \left(-\frac{8}{4}\right) R_2$

Now For $F^{(2)}$,

$$F^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

$m = -\frac{8}{4}$

$= -2$

So, $-m = 2$

So, Now,

$$[U] = A^{(3)} = F^{(2)} A^{(2)}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 8 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$ ← Upper Triangular Matrix

Now, Lower Triangular Matrix

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ m_1 & 1 & 0 \\ m_2 & m_3 & 0 \end{bmatrix}$$ Here Non negative val of f

Now for the eqn,

$$[L] \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$ ← Values of y

So,

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$$

So, $a_1 = 0$

$a_1 + a_2 = 4$; $a_2 = 4 - a_1 = 4 - 0 = 4$

$2a_1 - 2a_2 + a_3 = 4$; $a_3 = 4 - 2a_1 + 2a_2 = 4 - 0 + 2(4)$

$= 12$

So, $a_1 = 0$; $a_2 = 4$; $a_3 = 12$

Now,

$$[u] + \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 12 \end{bmatrix}$$

So,

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 12 \end{bmatrix}$$

So, $-2x_3 = 12$; $x_3 = -6$

$-4x_2 + x_3 = 4$; $x_2 = \dfrac{4 - x_3}{-4} = \dfrac{4 - (-6)}{-4}$

$= -2.5$

$x_1 + 2x_2 + x_3 = 0$

$\therefore x_1 = -x_3 - 2x_2 = -6 - 2(-2.5) =$

$= 11$

So, $x_1 = 11$; $x_2 = -2.5$; $x_3 = -6$

(Ans.)

$2x + y + z = 10$

$3x + 2y + 3z = 18$

$x + 4y + 9z = 16$

The Coefficient Matrix,

$$A^1 = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$$

Here,

$R_2' = R_2 - \left(\frac{3}{2}\right)R_1$; $m = 1.5$

$R_3' = R_3 - \left(\frac{1}{2}\right)R_1$; $m = 0.5$

$$\text{So, } F^1 = \begin{bmatrix} 1 & 0 & 0 \\ -1.5 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } A^2 = F^1 A = \begin{bmatrix} 1 & 0 & 0 \\ -1.5 & 1 & 0 \\ -0.5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 3.5 & 8.5 \end{bmatrix}$$

$R_3' = R_3 - \left(\frac{3.5}{0.5}\right)R_2$

$m = 7$

$$\text{Now, } F^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix}$$

Now

$$[u] = A^3 = F^2A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 3.5 & 8.5 \end{bmatrix}$$

$$\text{So, } [u] = A^3 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\text{So, } [L] = \begin{bmatrix} 1 & 0 & 0 \\ 1.5 & 1 & 0 \\ 0.5 & 7 & 1 \end{bmatrix}$$

Now,

$$\begin{bmatrix} L \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$a = 10,\ b = 3,\ c = -10$$

$$\text{Now, } U \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 0.5 & 1.5 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ -10 \end{bmatrix}$$

$$z = 5,\ y = 9,\ x = 7$$